



Battery Pack Balanced CELL CHARGE MONITORING SYSTEM

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Balanced charging of each cell in a battery pack is critical to meeting system requirements and maximising lifespan, while continuous monitoring of temperature, current, and charge level ensures safe, efficient operation. Although conventional chargers and BMS modules provide basic balancing and protection, they often fall short of delivering real-time wireless monitoring for remote battery status and health tracking.

This IoT-based balanced cell charger and monitoring system is a compact, real-time battery management solution for multi-cell Li-ion/LiPo packs ranging from 1S to 4S. It accurately monitors individual cell voltages, per-cell temperature, pack current and power, state of charge (SoC), and overall pack health. Support for 1S to 4S packs means the system can handle battery packs with 1 to 4 cells connected in series, covering an approximate voltage range of 3.7V to 14.8V for Li-ion/LiPo cells.

All collected data is presented on a clear, responsive web dashboard featuring live per-cell line graphs, colour-coded status cards (green for full, red for critically low), and intuitive battery level indicators. The system also supports dynamic configuration for 1S to 4S packs, different battery chemistries,

and capacities, while operating seamlessly in both Station mode (connected to home Wi-Fi) and Access Point (AP) mode.

In this design, an additional voltage sensor and current sensor are placed at the battery's main terminal to measure overall pack voltage, charge level, load current during operation, and corresponding power consumption. Separate voltage sensors are also used for up to four cells, along with an NTC thermistor for each cell, enabling individual monitoring of cell voltages, charging behaviour, and cell temperature.

Fig. 1 shows the author's prototype under testing. The components required to build this system are listed in the Bill of Materials table.

Circuit and working

Fig. 2 shows the circuit diagram of the battery pack balanced-cell charge-monitoring system. It is built around the IndusBoard COIN V2, along

BILL OF MATERIALS

Name	Designator	Quantity
10k	R1-R5, R7, R9, R12	8
47k	R6, R8, R10, R11	4
10k NTC thermistor	S1, S2, S3, S4	4
INA219 current sensor	S5	1
IndusBoard COIN V2	U1	1
Multi-cell 2S/3S/4S Type-C	U2	1
4-cell 18650 battery holder	U3	1

EFY notes: As U2 and U3 are an external charger and battery holder case, respectively, they are not shown in the circuit

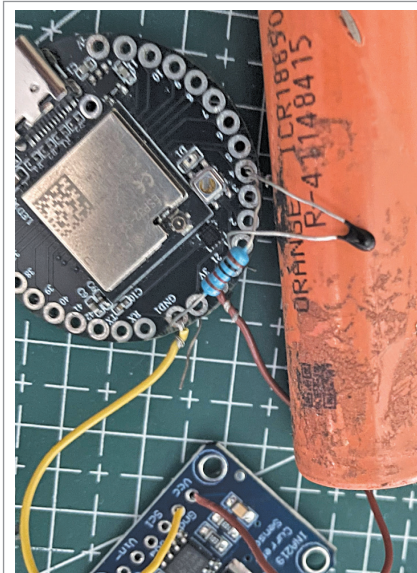


Fig. 1: Author's prototype used during testing

